

Physics 225, Spring 2026: Homework 2 - Solutions

Due 2/5/26

All solutions must clearly show the steps or reasoning you used to arrive at your result. You will lose points for poorly written solutions or incorrect reasoning/explanations; answers given without explanation will not be graded.

Name:

Section:

1. (50 points) **Fun with muons.** The muon is an unstable elementary particle with a half-life of $T_{1/2} = 3 \times 10^{-6}$ seconds, which means that if you have N muons at rest and wait 3×10^{-6} s, on average, half of them will have decayed and you will have $N/2$ remaining. However, the half-life is measured in units of *proper time*, which will differ from the time elapsed on the moving muon's clock as seen by a stationary observer.

- (a) (10 points) The muon storage ring at Fermilab has a diameter of 15 m and accelerates muons to a speed of $0.9994c$. If time dilation were *not* present, compute the number of revolutions around the ring the muons could make before half of them had decayed.

The time to circle the ring (ignoring time dilation) is $T = \frac{\pi \times 15 \text{ m}}{0.9994c} = 157 \text{ ns}$. The number of revolutions is $T_{1/2}/T \approx 19$.

- (b) (10 points) Now, properly taking into account time dilation, how many revolutions can the muons make before half have decayed? Note the difference! The experiment would be vastly less efficient were it not for special relativity which extends the lab-frame lifetimes of these short-lived particles.

The muon's gamma factor is $\gamma = \frac{1}{\sqrt{1-0.9994^2}} = 28.9$, so a lab observer sees the muon survive for a time $t_{lab} = \gamma T$. This means the number of revolutions is $28.9 \times 19 \approx 550$.

- (c) (20 points) Muons are created naturally on Earth by collisions of cosmic rays with the upper atmosphere. Consider a muon traveling at $0.9999c$ produced at a height of 20 km above the Earth's surface and traveling straight down. The probability of survival from the time of production, as measured by the muon's proper time, is $P(\tau) = 2^{-\tau/T_{1/2}}$. Calculate the probability that the muon survives to reach the Earth's surface. Perform this calculation in two ways: in the Earth frame, considering time dilation of the muon lifetime as in (b), but *also* in the moving muon's frame by considering length contraction. You should obtain the same answer in both cases: time dilation and length contraction are not really separate effects, but are two manifestations of the same property, namely the constancy of the speed of light.

A stationary observer will see the muon travel time as $t_{lab} = \frac{20 \text{ km}}{0.9999c} = 66.7 \text{ } \mu\text{s}$, but the muon's proper time is $\tau = t_{lab}/\gamma$, where $\gamma \approx 70.7$. The survival probability is then $P = 2^{-t_{lab}/(\gamma T_{1/2})}$. Another way to read that equation is that the muon's lab-frame lifetime is extended to be $\gamma T_{1/2}$. Plugging in numbers, we find $P \approx 80\%$. In the muon's frame, the Earth is moving up to meet the muon, and its distance away is length-contracted to $L = 20 \text{ km}/\gamma$. The Earth rushes up to meet the muon at a speed $0.9999c$, which takes a proper time $\tau = L/(0.9999c) = 944 \text{ ns}$. Then $P(\tau) = 2^{-\tau/T_{1/2}} = 80\%$, as before.

- (d) (10 points) Finally, as in part (a), compute the survival probability if relativistic effects were *not* present. The fact that we see any muons at all (as you can see for yourself in PHYS 403!) is everyday evidence for special relativity and time dilation.

Without relativistic effects, we would have found $P = 2^{-t_{lab}/T_{1/2}} = 2 \times 10^{-7}$!

2. (50 points) **Walking on a train.** Morin exercise 1.26 (p. 49).

To set up this problem, it's useful to consider two spacetime events: event A is “front of train crosses back of tunnel,” and event B is “back of train crosses front of tunnel.” The front and back of the tunnel are specific points in space in the ground frame (likewise with the front and back of the train in the train frame), and the crossings occur at specific times, so all observers can agree on the coordinates of events A and B as measured in different frames. Since “person begins walking” and “person ends walking” occur at the same time and place as events A and B, respectively, so they can be identified with events A and B.

- (a) We are asked for the time between events A and B as measured in the ground frame. In the ground frame, the tunnel has length L but the train gets length-contracted down to L/γ_{train} , where $\gamma_{train} = \frac{1}{\sqrt{1-(3/5)^2}} = 5/4$. So the total length that the train has to travel is the length of the tunnel plus the length of the train (both measured in the ground frame), $L_{tot} = L + \frac{4}{5}L = \frac{9}{5}L$. It travels this distance at speed $3c/5$ in the ground frame, so the travel time is $t_{ground} = \frac{9L/5}{3c/5} = \frac{3L}{c}$.
- (b) In the ground frame, the person walking has to traverse the length of the tunnel, which has length L . Doing so in a time t_{ground} gives speed $v_{person} = \frac{L}{t_{ground}} = \frac{c}{3}$. Note that because we are measuring both length and time in the ground frame means that we don't have to use the velocity addition formula.
- (c) We are now asked for the time between events A and B as measured in the frame of the person. Since the person is not moving in the person's frame, the desired time is the person's proper time τ , which is time-dilated from t_{ground} . The person's gamma factor is $\gamma_{person} = \frac{1}{1-(1/3)^2} = \frac{3}{2\sqrt{2}}$, so $\tau = \frac{t_{ground}}{\gamma_{person}} = 2\sqrt{2}\frac{L}{c}$.